



Linux day 2025

\$ whoami



Leonardo Tamiano

\$ groups

Computer Scientist, Programmer

Penetration Tester, Trainer,

Teacher



Esadecimale

@esadecimale · 7640 iscritti · 322 video

<https://esadecimale.it/>



L'intento è quello di mostrare l'essenza dell'informatica, nella sua sostanza più cruda, con ...altro

esadecimale.it e 3 altri link

Personalizza canale

Gestire i video



hexdump

@hexdump1337 · 26.100 iscritti · 184 video

Hello there! ...altro

patreon.com/hexdump e 2 altri link

Personalizza canale

Gestire i video

<https://hexdump.sh/>

Dead Internet Theory





Dead Internet Theory: Most of the Internet is Fake

 **IlluminatiPirate** ·  Jan 5, 2021 ·  **algorithm-driven behavioral loops**

dead internet theory

demoralization

homogenization

npcs

This thread has been viewed 362295 times.

<https://forum.agoraroad.com/index.php?threads/dead-internet-theory-most-of-the-internet-is-fake.3011/>



Maybe You Missed It, but the Internet ‘Died’ Five Years Ago

A conspiracy theory spreading online says the whole internet is now fake. It’s ridiculous, but possibly not *that* ridiculous?

By Kaitlyn Tiffany

<https://www.theatlantic.com/technology/archive/2021/08/dead-internet-theory-wrong-but-feels-true/619937/>

1. Human activity has been displaced by **bots** and **algorithmically curated search results**.





1. Human activity has been displaced by **bots** and **algorithmically curated search results**.
2. **state actors** are doing this in a coordinated effort to **manipulate the human population**.

1.



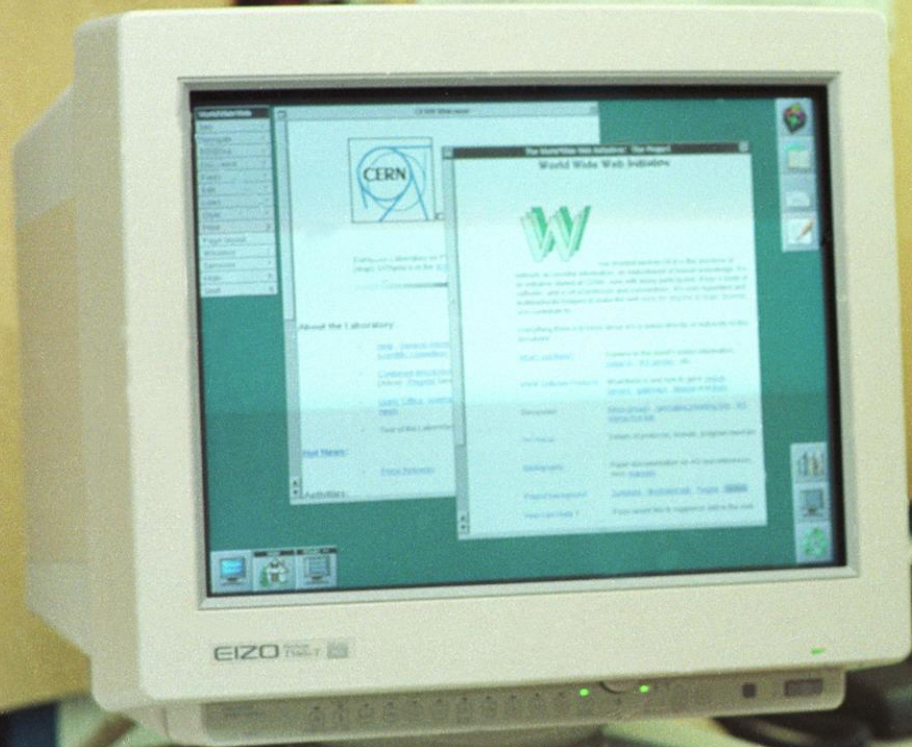
2.

AI

≠

**How did we
get here?**

Tim Berners-Lee,
1989, CERN.



Information Management: A Proposal



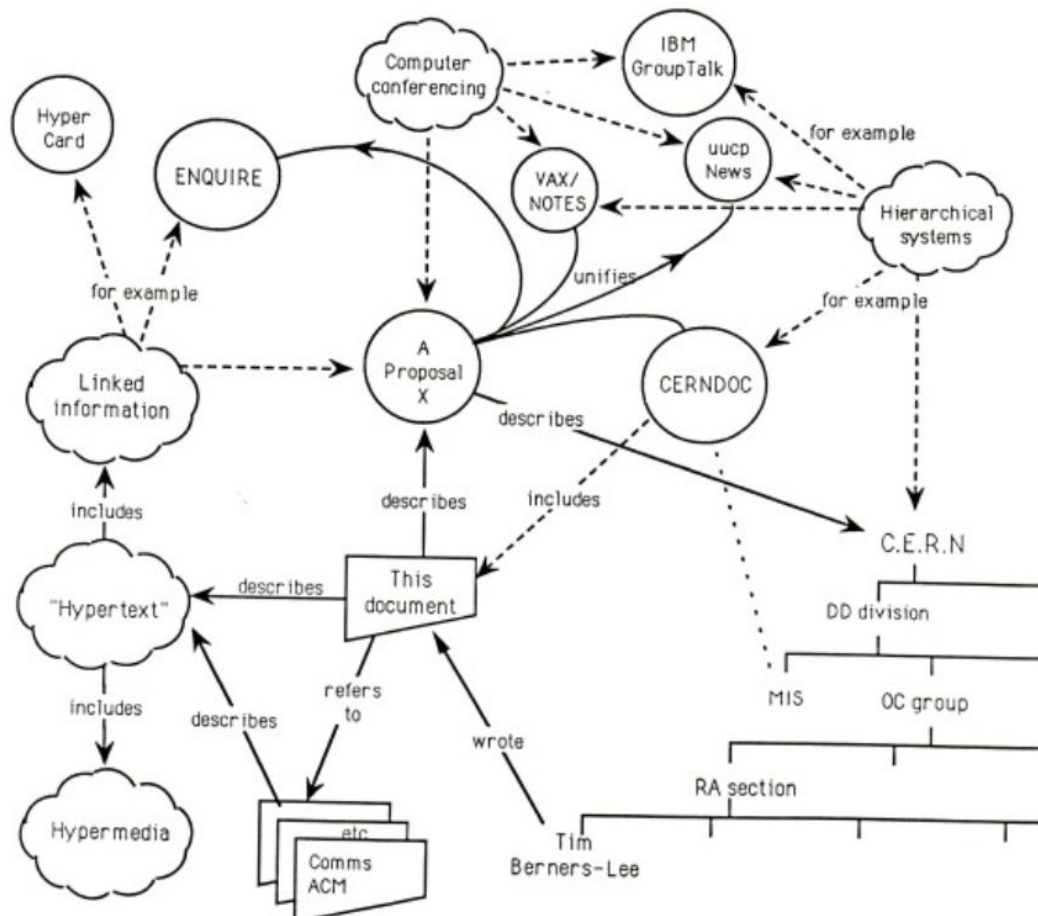
Tim Berners-Lee, CERN

March 1989, May 1990

This proposal concerns the management of general information about accelerators and experiments at CERN. It discusses the problems of loss of information about complex evolving systems and derives a solution based on a distributed hypertext system.

<https://home.cern/science/computing/birth-web/short-history-web>

This p
at CEI
derives

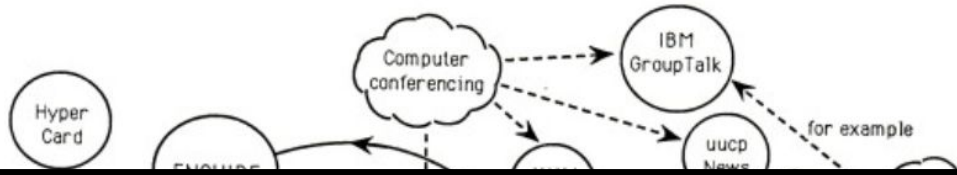


1

≠

rs and experiments
lving systems and

veb

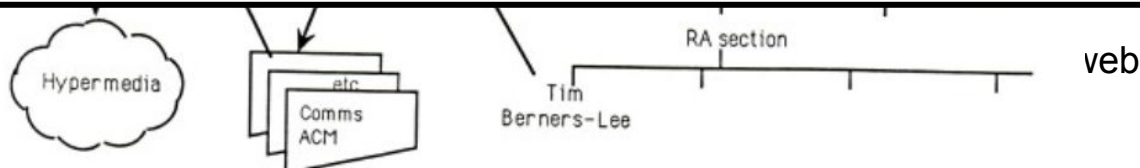


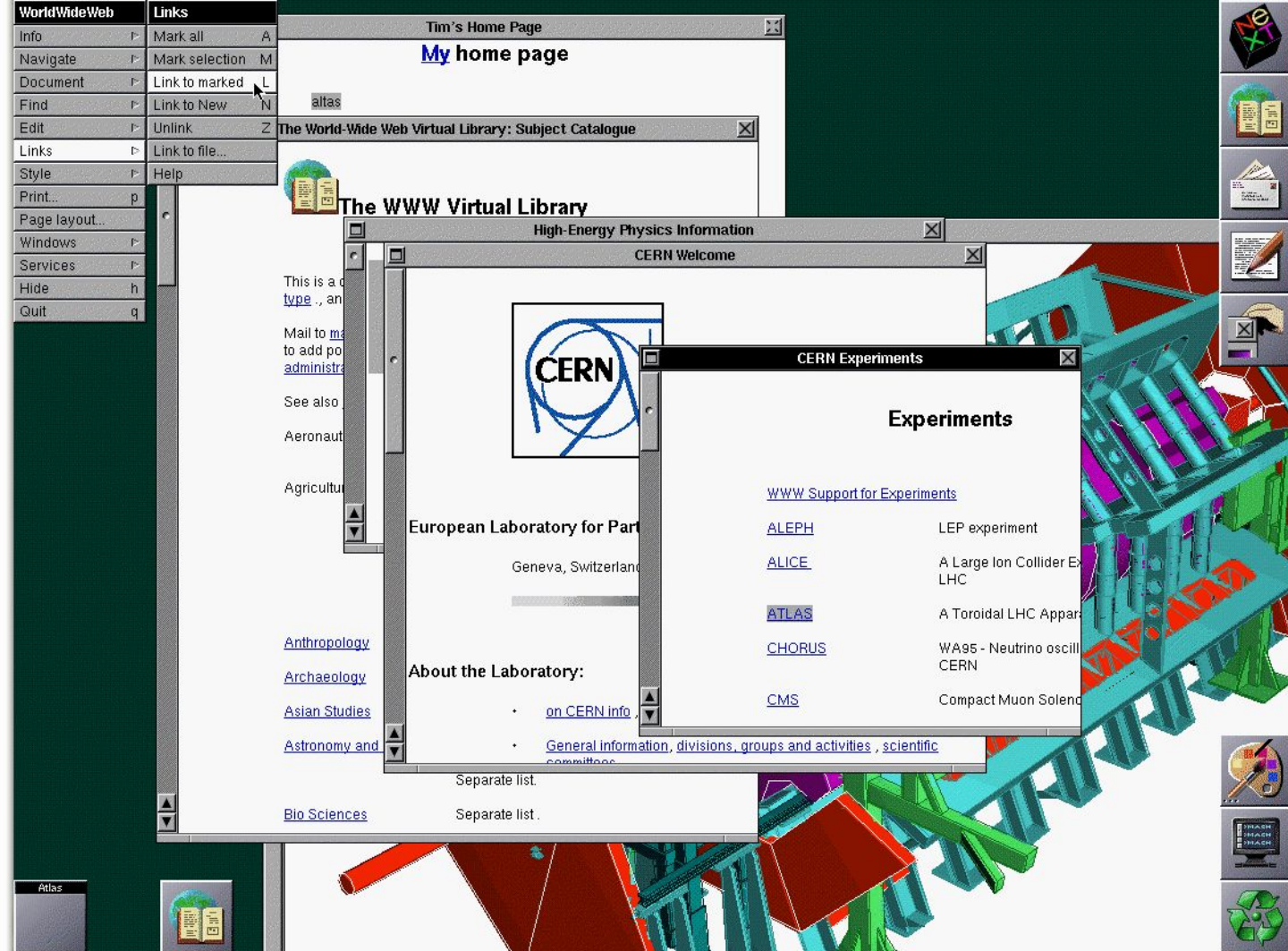
1



This proposal concerns the management of general information about accelerators and experiments at CERN. It discusses **the problems of loss of information about complex evolving systems** and derives a solution based on a distributed hypertext system.

This
at C
deri





1993,
NeXT
computer

Line Mode Browser (1991)



ViolaWWW (1992)

Mosaic (1993)

Netscape Navigator (1993)

Line Mode Browser (1991)



ViolaWWW (1992)

Mosaic (1993)

Netscape Navigator (1993)



- **HyperText Markup Language (HTML)**



- **Cascading Style Sheets (CSS)**

- **JavaScript (JS)**

2 CA-2000-02: Malicious HTML Tags Embedded in Client Web Requests



This advisory is being published jointly by the CERT Coordination Center, DoD-CERT, the DoD Joint Task Force for Computer Network Defense (JTF-CND), the Federal Computer Incident Response Capability (FedCIRC), and the National Infrastructure Protection Center (NIPC).

Original release date: February 2, 2000

Last revised: February 3, 2000

<https://www.sei.cmu.edu/library/2000-cert-advisories/>

A complete revision history is at the end of this file.



Losing ground



- My Profile [edit]
- My Friends
- My Groups
- My Parties
- My Messages
- My Account
- My Privacy

Summer Jewish Adventure
New York City



```

00000001 11 10 1000
00000000 01 0 1
00000000 110 0 1100
00000000 010 010 101
00000000 001 00 01
00000000 0 01 010
00000000 1010 1 10 0
  
```

Brian Moore's Profile

Puget Sound

Picture



[Send Brian a Message](#)

[Poke Him!](#)

Connection

You are in a relationship with Brian.

Mutual Friends

You have 19 friends in common with Brian.

Access

Brian is currently logged in from a non-residential location.

Friends at Puget Sound



Kerala Hise



Jennifer McFadden



Amy Graeber



Sarah Spurlock



Vu Tran



Briana Skirm

Information

Account Info:

Name: Brian Moore
Member Since: May 21, 2005
Last Update: July 19, 2005

Basic Info:

School: Puget Sound '09
Status: Student
Sex: Male
Residence: Todd 311
Birthday: 09/02/1986
Home Town: Shorewood, WI 53211
High School: Shorewood Hi '05

Contact Info:

Email: bmoore@ups.edu
Screenname: Doctabu
Mobile: 414.702.7426
Websites: <http://www.doctabu.com>
<http://www.livejournal.com/users/doctabu>
<http://www.flickr.com/photos/doctabu>

Personal Info:

Looking For: Friendship
Interested In: Women
Relationship Status: In a Relationship with Rachel Buethe (Tiny Tykes Day Care)

Political Views: Very Liberal

Interests: Film, Graphic Design, Video Editing, Computers, Bowling, Dancing, Acting, Singing, Listening to Decent Music, Sleeping, Being Crazy

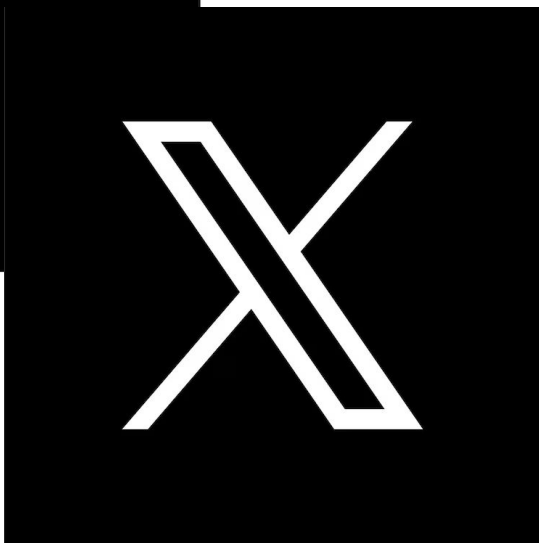
Favorite Music: Beck, The Beatles, They Might Be Giants, Phoenix, Paul Simon, OutKast, Avenue Q, Radiohead, Red Hot Chili Peppers, Steely Dan, Styx, U2, Zero 7

Favorite Movies: Office Space, Garden State, Eternal Sunshine of the Spotless Mind, Triplets of Belleville, Amelie

About Me: What is there to say? I dunno. I'm a wild 5'10" skinny-ass that wants to be creative. I like making movies and stuff like that.







Problems of **using ONLY big platforms**



- Cannot **modify** the look of the platform.
- Sharing becomes a **commodity**.
- Limited **active drive**.
- Lots of **passive consumption**.

Problems of **ONLY** using big platforms



We become **consumers** of content

Problems of **ONLY** using big platforms



- We lose our ability to **select**

- We become **consumers** of content



Problems of **ONLY** using big platforms



- We lose our ability to **select**
- We become **consumers** of content
- We lose our **attention**

Our need for **community** has
been **commercialized**



Our need for **community** has
been **commercialized**

≠



NordVPN®

Our need for **community** has
been **commercialized**



Abbiamo veramente bisogno di una VPN?



Esadecimale • 4688 visualizzazioni • 1 anno fa

L'obiettivo di questo video è quello di fare chiarezza sul perché le VPN
non vanno ad aumentare la sicurezza dei loro utenti. Non è mia...

≠

A man and a woman are seen from behind, standing in a high-rise office at night. They are holding hands and looking out a large window at a city skyline with illuminated buildings. The office interior is dimly lit, with a desk and chair visible on the left and a lamp on the right.

You met me at a very strange time in my life.

Self-Host!



≠

- Get a **VPS**
- Install **Linux**
- Login with **SSH**
- Get a **domain** name
- Self-Host!



VPS -> Virtual Private Server



A machine on the internet!

- Good **bandwidth**.
- Ok **CPU** and **Disk**
- Meh **RAM** (unless you pay!)

Linux

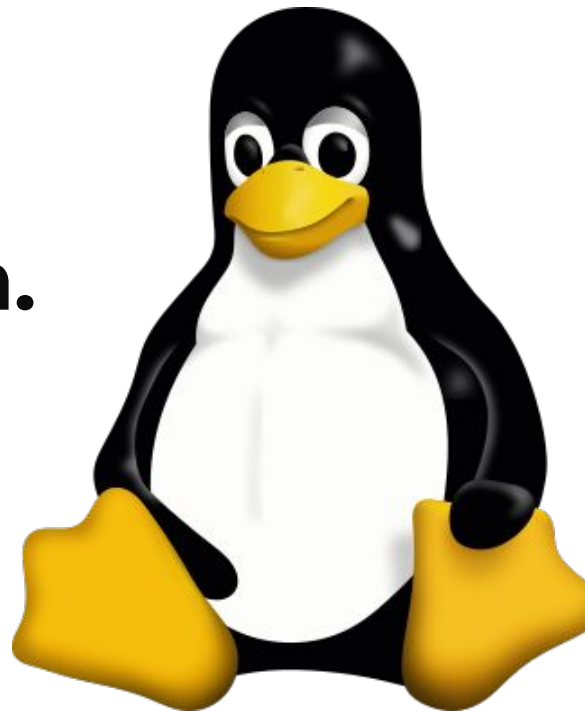


It's an Operating System.

You can view its source-code.

You can use it.

It tries to protect your freedom.



Programmazione del Kernel Linux con QEMU

35:38

Programmazione del Kernel Linux con QEMU

Esadecimale • 1217 visualizzazioni • 1 anno fa

Ciao! In questo video mostro un setup basato su initramfs e QEMU per andare a giocare con i moduli del kernel linux. L'idea quindi è quella di...



User space Kernel space

 Esadecimale

30:52

Modifichiamo il Kernel Linux!

Esadecimale • 2488 visualizzazioni • 4 mesi fa

Iscriviti al patron per accedere al materiale dei video! -
<https://git.hexdump.sh> I miei progetti: - <https://esadecimale.it> - ...



SSH -> Secure Socket Shell



- A **protocol** to talk to our VPS.
- Securely, thanks to **cryptography**.
- No need for paid VPNs.

```
ssh user@hexdump.sh
```



```
ssh -p1337 user@hexdump.sh
```

```
ssh -i id_rsa -p1337  
user@hexdump.sh
```

DNS (Domain Name System)



Protocol that translates domains into IP addresses.



DNS (Domain Name System)



*It's not DNS
There's no way it's DNS
It was DNS*

-SSBroski





Level 0
Web Page

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>A Simple Page</title>
</head>
<body>
  <header>
    <h1>Welcome to A Simple Page</h1>
  </header>
  <main>
    <p>Let's get back our power!</p>
  </main>
</body>
</html>
```



GitHub static pages



- Create new repository
- Name it “**username.github.io**”
- Make it public
- Push the content
- Wait for profit





leonardoe95.github.io

Public



Pin



Watch

0



m... ▾



1 Branch



0 Tags



Go to file



<> Code ▾



LeonardoE95 First ✓

4828d2d · 4 minutes ago



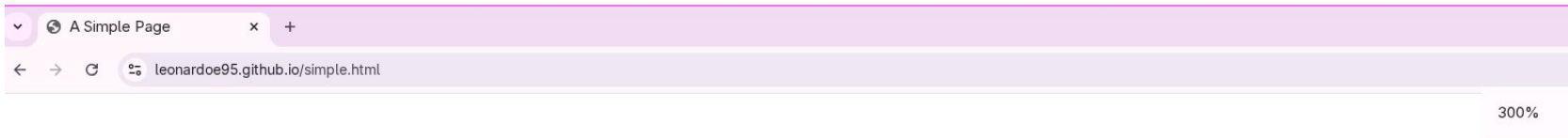
1 Commit



simple.html

First

4 minutes ago



Welcome to A Simple Page

Let's get back our power!

<https://leonardoe95.github.io/simple.html>

<https://motherfuckingwebsite.com/>



← → ↻ 🌐 motherfuckingwebsite.com

This is a motherfucking website.

And it's fucking perfect.

Seriously, what the fuck else do you want?



Level 1

Web App

Different options for **hosting**:



- GitHub static pages
- Private VPS on the cloud
- Powerful server on home network
- Small device on home network (raspberry)

Private VPS on the cloud

- Configure SSH access
- Install web server
- Configure virtual host
- Configure DNS
- Obtain TLS certificate
- Upload website



**Virtual
Private
Server**

DNS record A



esadecimale.it



84.234.16.47

DNS record CNAME

esadecimale.it



hexdump.sh

Make sure you can **ping the host**.



```
[leo@work data]$ ping -c2 esadecimale.it
PING esadecimale.it (84.234.16.47) 56(84) bytes of data.
64 bytes from ov-90467b.infomaniak.ch (84.234.16.47): icmp_seq=1 ttl=46 time=35.3 ms
64 bytes from ov-90467b.infomaniak.ch (84.234.16.47): icmp_seq=2 ttl=46 time=35.2 ms

--- esadecimale.it ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 35.183/35.232/35.281/0.049 ms
```

Install web server

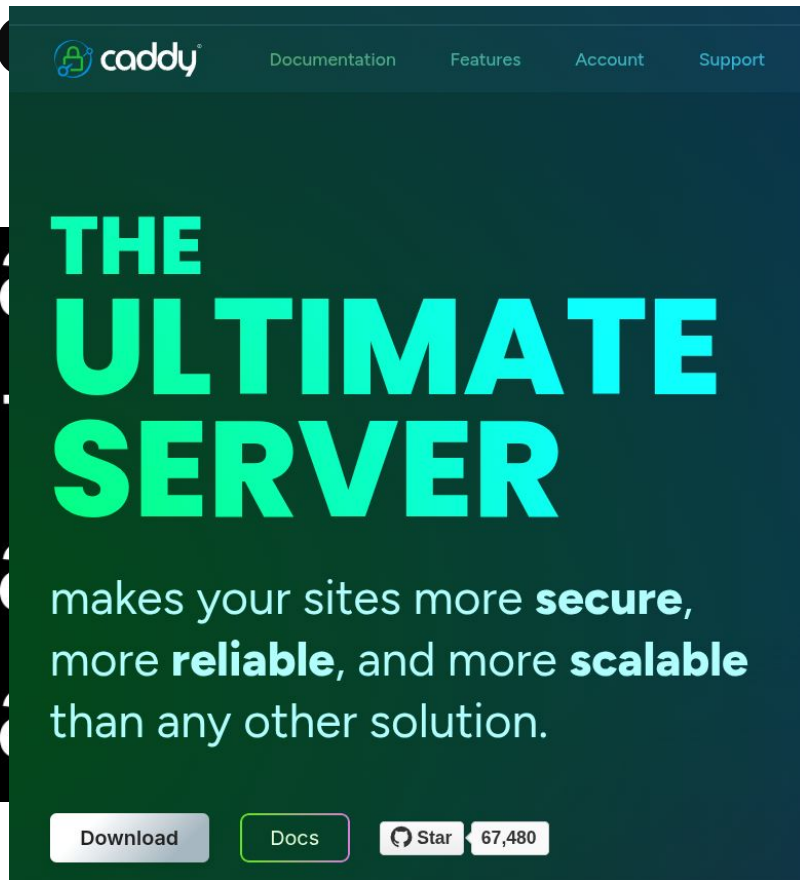


```
apt-get update  
apt-get install caddy  
systemctl caddy enable  
systemctl caddy start
```

Install web server

<https://caddyserver.com/>

```
apt-get update  
apt-get install  
systemctl ca  
systemctl ca
```



The screenshot shows the Caddy website homepage. The header is dark blue with the Caddy logo (a padlock icon) and the word "caddy" in white. To the right of the logo are four links: "Documentation", "Features", "Account", and "Support". The main content area has a dark green background. The title "THE ULTIMATE SERVER" is written in large, bold, cyan letters. Below the title, the text "makes your sites more **secure**, more **reliable**, and more **scalable** than any other solution." is displayed in white. At the bottom, there are three buttons: "Download" (grey), "Docs" (green outline), and a GitHub Star button showing "67,480" stars.

 caddy

[Documentation](#) [Features](#) [Account](#) [Support](#)

THE ULTIMATE SERVER

makes your sites more **secure**, more **reliable**, and more **scalable** than any other solution.

[Download](#) [Docs](#)  Star 67,480



Upload website data



```
tar -czvf data.tar.gz /data  
scp data.tar.gz user@vps:/remote/path
```

/etc/caddy/Caddyfile

≠

```
esadecimale.it {  
    root * /var/www/html/esadecimale.it  
    file_server  
}
```



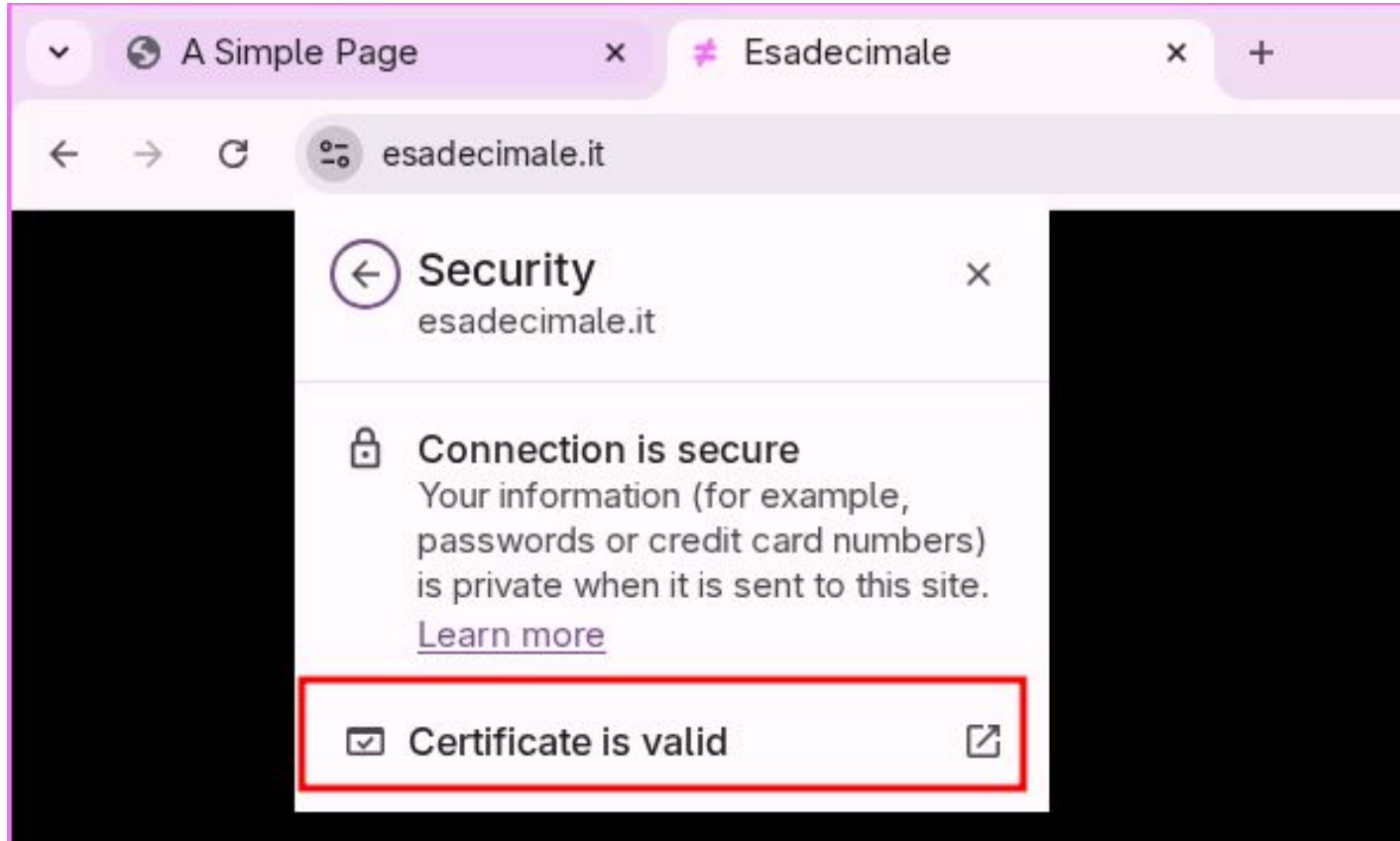
```
caddy fmt --overwrite  
/etc/caddy/Caddyfile
```



```
caddy validate --config  
/etc/caddy/Caddyfile
```

```
systemctl reload caddy
```

Obtain a **TLS certificate**.

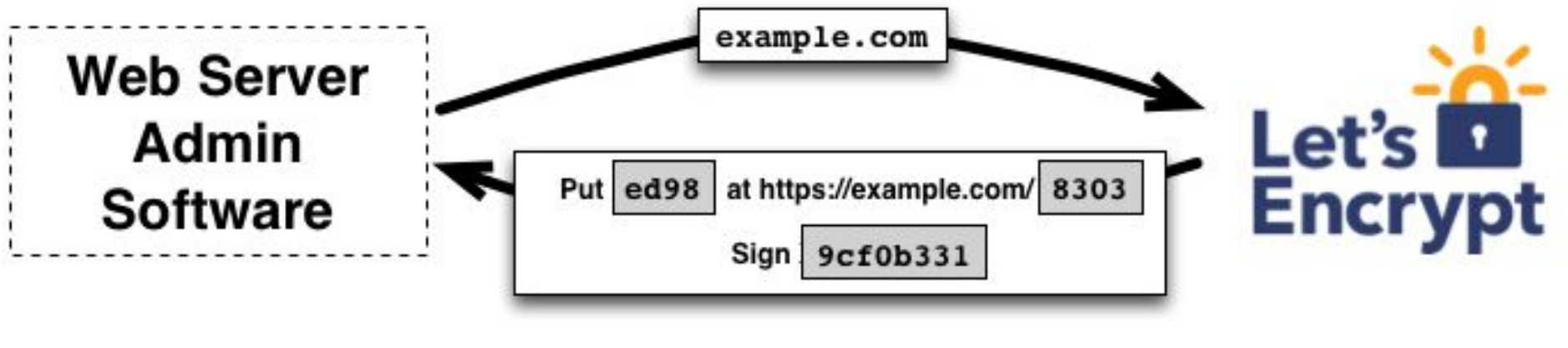




Encryption for Everybody

A nonprofit providing free TLS certificates to more than **700M** websites.

[Get Started](#)[Sponsor](#)



to more than 700M websites.

[Get Started](#)[Sponsor](#)

informatica.uniroma2.it ×

⚠ Your connection to this site is not secure
You should not enter any sensitive information on this site (for example, passwords or credit cards), because it could be stolen by attackers. [Learn more](#)

🍪 Cookies and site data >

⚙ Site settings ↗

INFORMATICA TRIENNALE

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Magistrale

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Corso di laurea triennale in Informatica

Università degli studi di Roma "Tor Vergata"



Primo piano



AVVISO - Spostamento Aule 28 e 29 ottobre 2025

23-10-2025 14:59 inviato da Stillitano

Si informano gli studenti che alcune lezioni subiranno variazioni di aula nei giorni martedì 28 e mercoledì 29 ottobre 2025.

📅 Martedì 28 ottobre 2025

- Programmazione dei calcolatori con laboratorio – Prof. Gianluca Rossi
🕒 09:00–11:00 → Aula 6 PP2
- Analisi matematica – Prof. Roberto Tauraso
🕒 11:00–13:00 → Aula 2 PP2



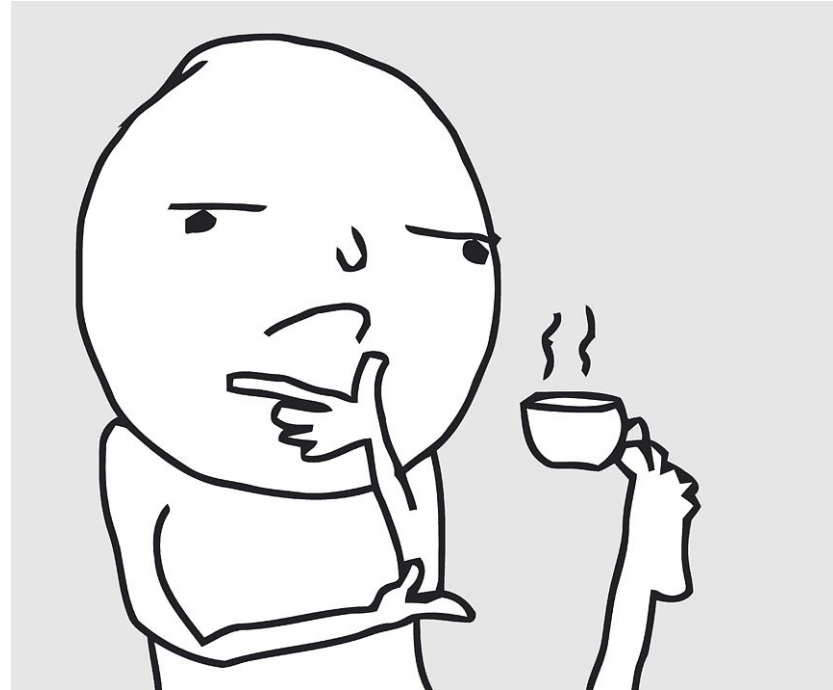
You want to host a web application.



This application only comes with **HTTP**.

No TLS supported is provided.

What do you do?



You want to host a web application.

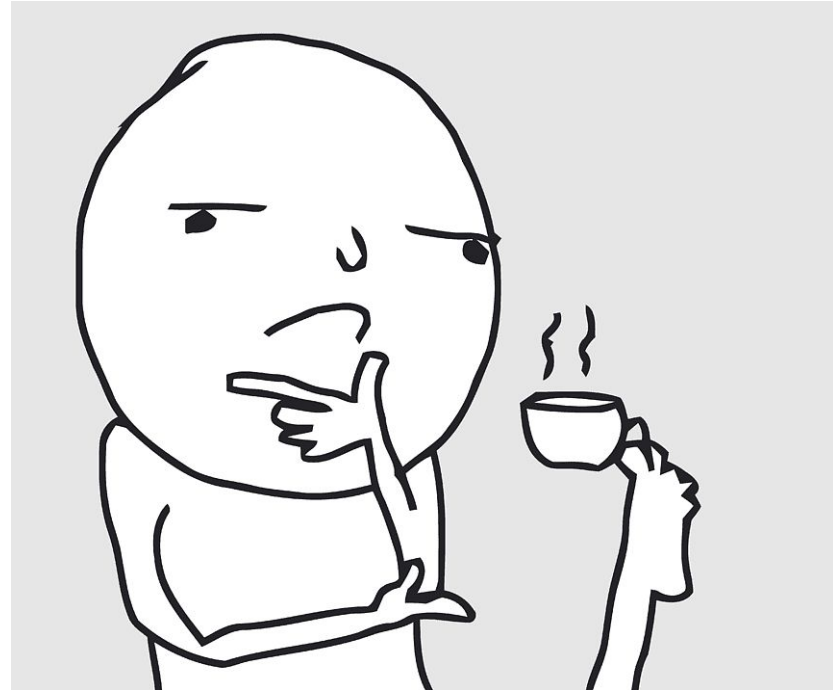


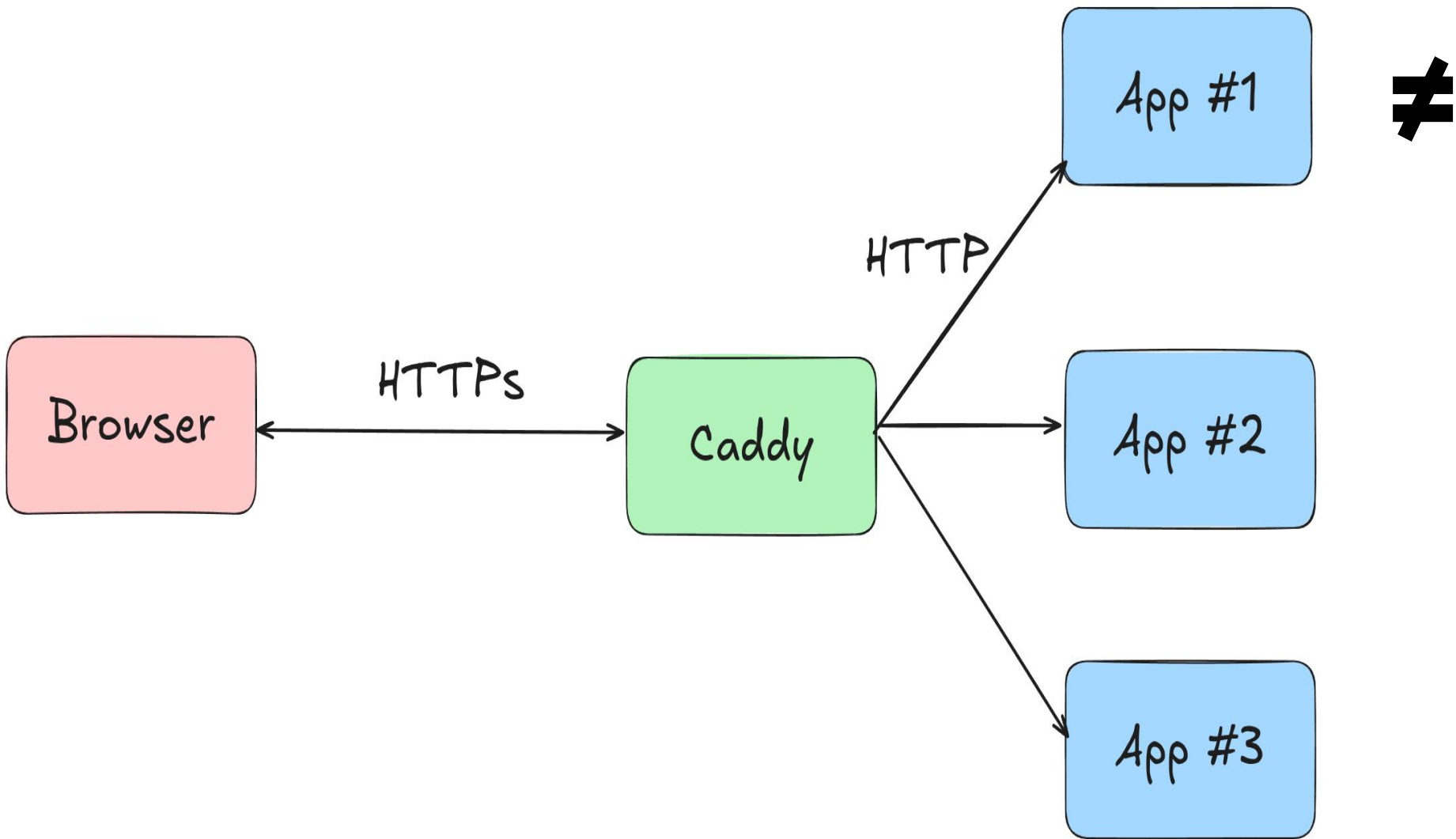
This application only comes with **HTTP**.

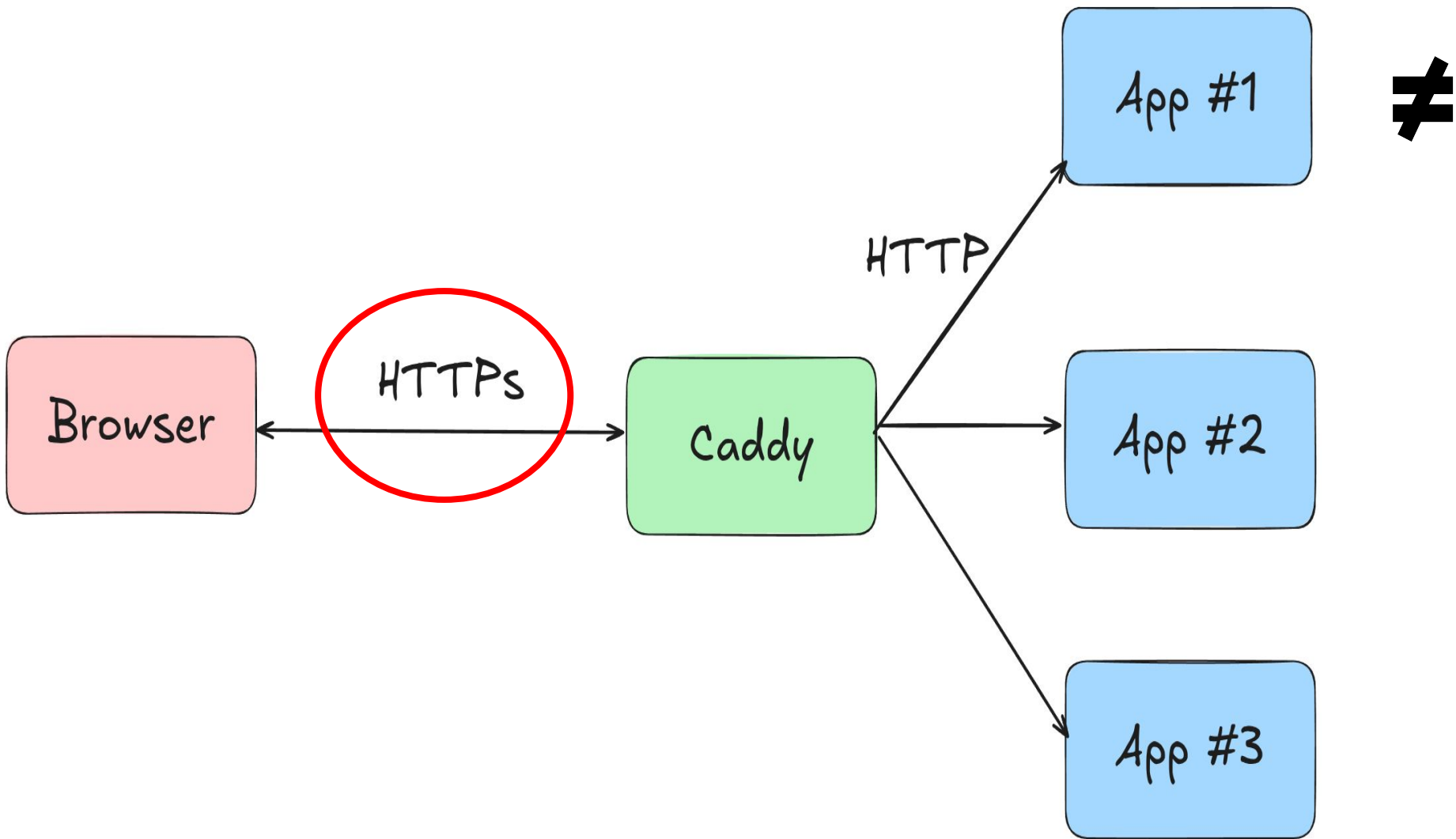
No TLS supported is provided.

What do you do?

Reverse Proxy!









**CHOOSE CLARITY. NOT
CHAOS. CHOOSE
PORTAINER.**

Create docker volume

```
#+begin_src sh
```

```
docker volume create portainer_data
```

```
#+end_src
```

Download and run container with simple HTTP port

```
#+begin_src sh
```

```
docker run -d \
```

```
    -p 1112:9000 \
```

```
    --name portainer \
```

```
    --restart=always \
```

```
    -v /var/run/docker.sock:/var/run/docker.sock \
```

```
    -v portainer_data:/data \
```

```
    portainer/portainer-ce:latest
```

```
#+end_src
```



/etc/caddy/Caddyfile



```
portainer.hexdump.sh {  
    reverse_proxy 127.0.0.1:1112 {  
        header_up Connection {>Connection}  
        header_up Host {host}  
        header_up X-Real-IP {remote_host}  
    }  
}
```

/etc/caddy/Caddyfile



```
portainer.hexdump.sh {  
    reverse_proxy 127.0.0.1:1112  
}
```

Btw, sometimes **ssh port forwarding** is useful.



```
ssh  
-L 1337:127.0.0.1:1112  
user@admin.sh
```

Btw, sometimes **ssh port**

≠

for **ssh(1) - Linux man page**

Name

ssh - OpenSSH SSH client (remote login program)

`user@admin.sh`



Self-host Whatever

- Find **cool projects**
- Read **documentation**
- Deploy and have fun



Level 2

Freelancing

2025
05
01



Mi sono licenziato. Parliamone.

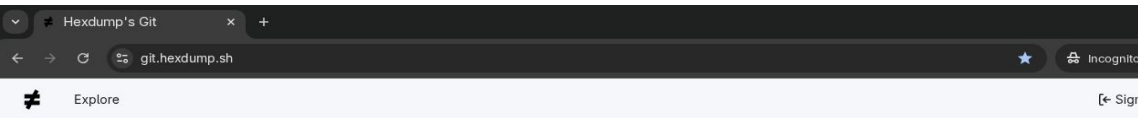


4491 visualizzazioni • 4 mesi fa

https://esadecimale.it



https://git.hexdump.sh



Hexdump's Git



Dotfiles

Configuration files for the software I use daily, such as Emacs, i3, BurpSuite.



Study notes

Notes related to everything I study for my own personal learning and for my YouTube content.



Code

Scripts, programs and PoCs
own personal learning and
content



Certification Material

Notes, cheatsheets, checklists, programs and
other material aimed at preparing for certifications
in the context of offensive security.



[hexdump / kernel-dev](#)

Updated last week



[hexdump / youtube.el](#)

Private

Emacs-Lisp automation for managing my YouTube accounts.

Updated last week



[hexdump / wapt.el](#)

Updated 2 weeks ago

<https://patreon.com/hexdump>



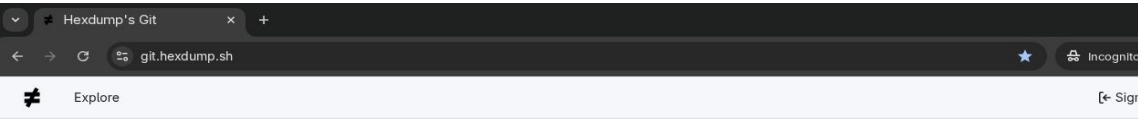
**Cybersecurity
Programming
Computer Science**



Hexdump

High-Quality education on Cybersecurity and Programming

https://about.gitea.com/

[Products](#) ▾[Resources](#) ▾[Community](#) ▾[Pricing](#)[Cloud](#)[Sign In](#)[Contact Us](#)

Private, Fast, Reliable DevOps Platform

Great self-hosted option that brings teams and developers high-efficiency, but
easy operations from planning to production.



Study notes

Notes related to everything I study for my own
personal learning and for my YouTube content.



Certification Material

Notes, cheatsheets, checklists, programs and
other material aimed at preparing for certifications
in the context of offensive security.

https://forum.esadecimale.it



Better Software Conference

■ Programmazione



0

Practical Web Application Penetration Test (PWAPT)

■ Cybersecurity corso



0

Sulla Legacy di DOOM

■ Game Development



0

Mappare i dati usati dalle IA: perché è fondamentale e come possiamo farlo

■ Community



2

GitHub Disclaimer

■ Cybersecurity



4

[TECH ANALYSIS] CVE-2025-37899 – Esempio Base di Sfruttamento UAF in ksmbd (Linux Kernel SMB)

■ Cybersecurity



0

https://forum.hexdump.sh

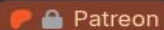


MALDEV 2 - NPM Crypto Malware Analysis



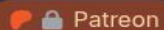
hexdump_t • Sep 15

MALDEV 1 - Introduction to Malware



hexdump_t • Sep 8

SCR 3 - SecureBank Code Review



hexdump_t • Sep 8

SCR 2 - A Practical Example in Java Spring



Sep 8

SCR 1 - Introduction to SCR



Sep 8



About ▾ Features Discover Enterprise Pricing Resources

Get Started

English



<https://www.discourse.org>

The online home for your community

A powerful platform for support, teamwork, and everything
community



<https://nextcloud.com/>

Regain control over your data

Remote collaboration made
easy



Leonardo Tamiano

Chiamata Conoscitiva

Ciao,

Qui puoi prenotare una chiamata gratuita della durata di 30 minuti al fine di conoscerci meglio e discutere insieme come posso aiutarti nel modo migliore possibile.

Ti aspetto!

Select a date

« < September 2025 > »

Su	Mo	Tu	We	Th	Fr	Sa
31	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30	1	2	3	4
5	6	7	8	9	10	11

Europe - Rome



Select slot

2:00 PM - 2:30 PM

2:30 PM - 3:00 PM

3:00 PM - 3:30 PM

3:30 PM - 4:00 PM

4:00 PM - 4:30 PM

4:30 PM - 5:00 PM





Type



Modified



People



Name ▲



Lecture-2025



Training-2025

Easy to use and privacy-friendly Google Analytics alternative



Plausible is powerful, intuitive and lightweight analytics. No cookies just insights.

Made and hosted in the EU, powered by European-owned infrastructure. 

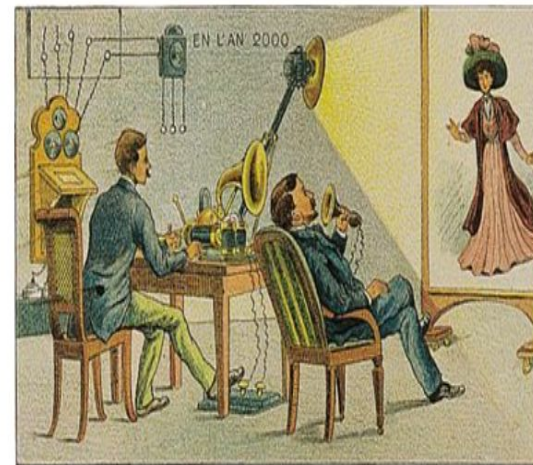
Get started

Live demo

<https://plausible.io/>

Galene videoconference server

Galene is a videoconference server that is easy to host and that requires very moderate server resources. It was originally designed for lectures, conferences and student tutorials, but later turned out to be useful for traditional meetings. *Galene* has been used in production at two major universities (*Université de Paris* and *Sorbonne Université*) for lectures, practicals, seminars, and for staff meetings. It has been used to host a number of conferences (including SOCS'2020, JFLA'2021 and LibrePlanet 2024).



<https://galene.org/>

```
#+begin_src sh
sudo docker run -it \
  -d --rm --name galene \
  -p 127.0.0.1:8443:8443 \
  -v $(pwd)/galene/groups:/opt/galene/groups \
  -e GALENE_GROUPS=/opt/galene/groups \
  -v $(pwd)/galene/passwd:/opt/galene/passwd \
  -e GALENE_PASSWD=/opt/galene/passwd \
  -v $(pwd)/galene/data:/opt/galene/data \
  deburau/galene /opt/galene/galene -insecure
#+end_src
```



galene + TURN server

```
#+begin_src sh
sudo apt update
sudo apt install coturn
#+end_src
```




Prometheus

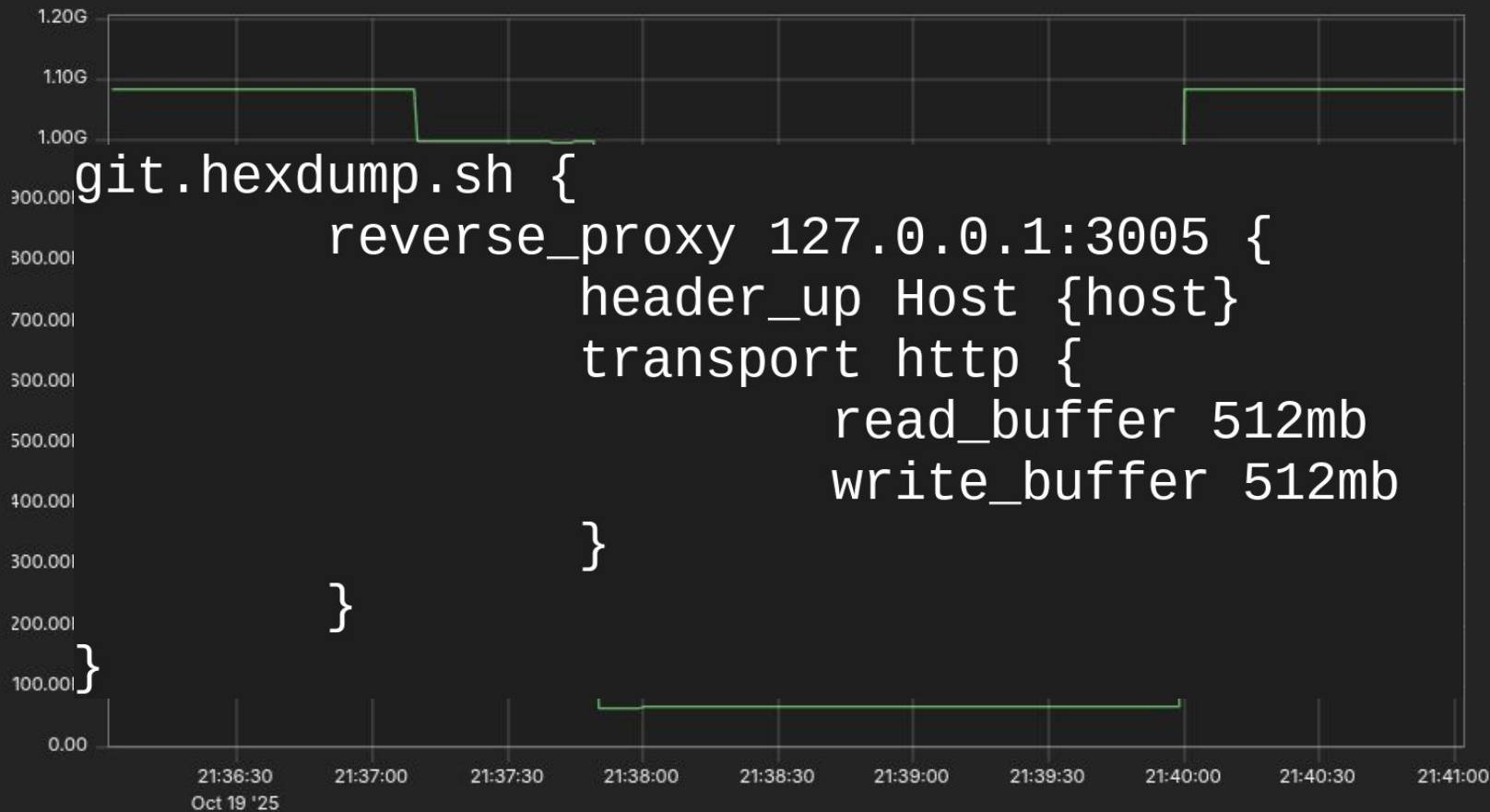


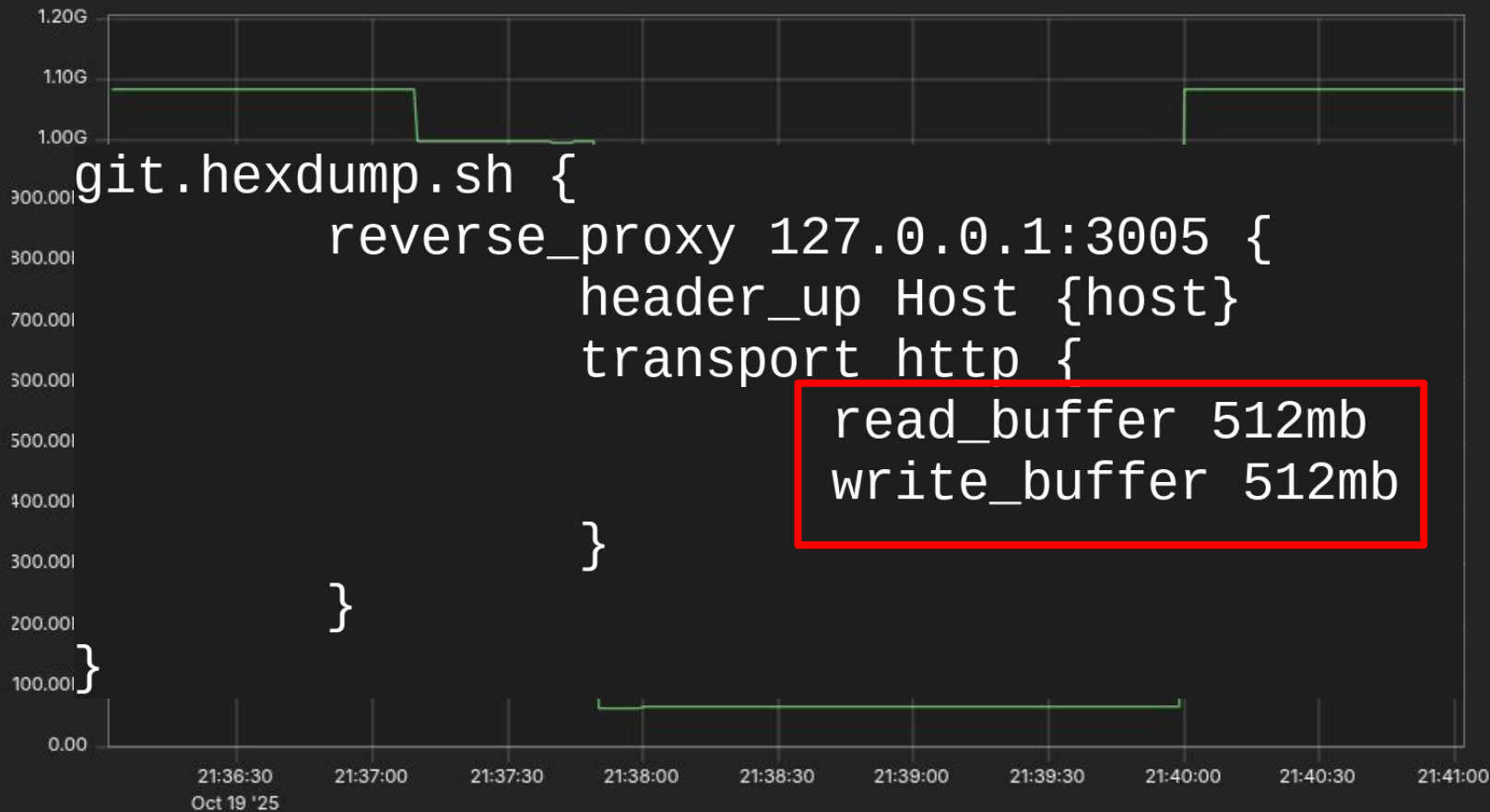
Open source **metrics and monitoring** for your systems and services

Monitor your applications, systems, and services with the leading open source monitoring solution. Instrument, collect, store, and query your metrics for alerting, dashboarding, and other use cases.



process_resident_memory_bytes(instance="localhost:2010", job="caddy")







Self-host in P.IVA

- Caddy
- Portainer
- Discourse
- Gitea
- Nextcloud
- Galene
- Plausible
- Prometheus
- **Bitwarden**
- ...



Level 3

?????????



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INSTRUCTOR



Leonardo Tamiano

Cybersecurity Trainer and Computer Science Teacher

<https://www.udemy.com/user/leonardo-tamiano-5/>

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Level 3

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